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Electron beam curing of acrylated epoxy resins for anisotropic conductive film application

Tae gyu Shin [a,1], Inhyuk Lee [a], Jungmin Lee [a], Jinyoung Hwang [b], Hoeil Chung [b], Kwanwoo Shin [c], Young soo Seo [d], Jaeyong Kim [a,*]

[a] Department of Physics and Institute of Nanoscience and Nanotechnology, Hanyang University, Seoul, 133-791, South Korea
[b] Department of Chemistry, Hanyang University, Seoul, 133-791, South Korea
[c] Department of Chemistry and Interdisciplinary Program of Integrated Biotechnology, Institute of Biological Interfaces, Sogang University, Seoul 121-742, South Korea
[d] Department of Nanoscience Technology, Sejong University, Seoul, 143-747, South Korea

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ABSTRACT

Radiation curable acrylated epoxy oligomer was found to be an effective resin system for application to electron beams (EB) on curing of anisotropic conducting film. To study curing degree as a function of EB dosage, we irradiated bisphenol-A type acrylate epoxy oligomer samples with doses of 5 to 600 kGy of EB. To investigate the effect of a metal barrier for potential industrial application, a 3 mm thick Al plate was placed in front of the samples, and the curing parameters were compared with the ones irradiated without an Al plate. As the dosage of the EB irradiation was increased, the glass transition temperature of the sample ranged from 46.8 to 62.2 °C for the epoxy composites without placing an Al plate, and from 46.4 to 64.1 °C for their counterparts with a 3 mm thick Al plate. These results confirm that enhancement of the curing degree with increasing EB irradiation is possible even in the presence of a metal plate. The scanning electron microscope images of the fracture surfaces are presented as evidence of the morphological changes of the EB cured epoxy samples.

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1. Introduction

As the density of integrated circuits is increasing, the demand for an innovative technology that can replace conventional soldering is also increasing. For display devices, the use of anisotropic conductive films (ACFs), which connect highly-integrated electric circuits and display panels has become a common practice to reduce toxicity, lead, and use of chlorofluorocarbon-based flux cleaners. ACFs are adhesive films with electrical conductivity along the out-of-plane direction and an insulating property along the in-plane direction. To exhibit its anisotropical electric properties, ACFs include micrometer size conductive particles in an epoxy resin having high cohesive and adhesive strength, low shrinkage, and versatility against external force [1,2].

It is well known that the properties of thermosetting epoxy resins critically depend on the curing conditions, including the curing temperature and time. Resins cured by electron beam (EB) irradiation exhibit many technical advantages for practical applications compared to thermally-cured and chemically-cured ones. These advantages include rapid curing at low temperature, reduced volatile emissions, improved resin stability, better material handling, reduced processing costs, and high shelf-life of materials [3–6]. One of the resins that are sensitive to EB irradiation, acrylate oligomer, undergoes a free-radical polymerization mechanism after being subjected to EB irradiation. The difference in the morphology of the EB irradiated oligomers can be attributed to the different degrees of curing of the epoxy and to the difference in the dose rate.

In this study, the EB irradiation dosage was varied for radiation curable acrylate epoxy oligomers which can be cured at room temperature with minimum irradiation time. The results of the morphological changes as well as the physical and mechanical properties of the ACF resin systems as a function of the EB irradiation are discussed. The influence of EB irradiation on the curing degree of the epoxy resin is analyzed by using a differential scanning calorimetry (DSC), a gel fraction test, Raman spectroscopy, and SEM images.

2. Experimental procedure

KDU651TP75, a bisphenol-A type acrylate epoxy oligomer (Kukdo Chemical Co. Ltd.) which is a mixture of 75% KDU-651 and 25% tripropylene glycol diacrylate (TPGDA) (see Fig. 1) was kindly supplied by Kukdo Chemicals, South Korea, and was used as received.

An appropriate amount of epoxy was transferred onto the slide glasses and covered with and without an Al plate. To maintain the electric connectivity after the curing of the epoxy by EB irradiation, the samples were uniformly pressurized by using an Al plate so that the conductive particles could touch the connecting circuits. In this sense, measuring the penetration depth of EB in an Al plate is

* Corresponding author. Tel.: +82 2 2220 0917; fax: +82 2 2295 6868.
E-mail address: kimjy@hanyang.ac.kr (J. Kim).
[1] Current address: Neutron Science Division, Korea Atomic Energy Research Institute, Daejeon, 305-353, South Korea.

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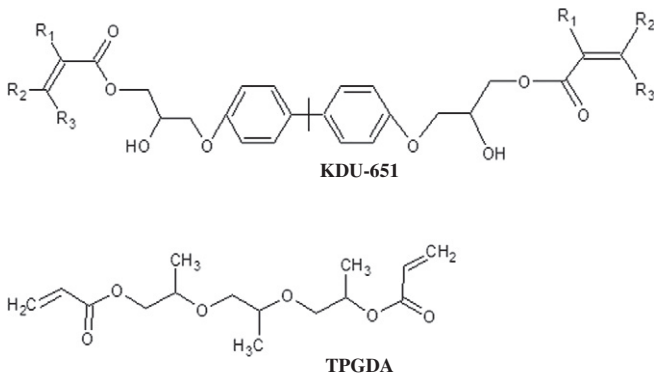

Fig. 1. Schematic diagrams of KDU-651 and TPGDA.

necessary. The samples were irradiated at the Advanced Radiation Technology Institute Accelerator facility in Jeongup, South Korea, using a 10 MeV accelerator (Model UELV-10-10S) at an EB current of 0.53 mA and 10 kW of power on average. With the constant scanning speed of the conveyer (0.1 m/s), this linear EB accelerator supplies an EB irradiation dose rate of 100 Gy/s. The epoxy samples were irradiated with doses of 5, 10, 20, 40, 80, 200, 400 and 600 kGy of EB. To investigate the effects of an Al plate, which is necessary to press the printed circuit board (PCB) plates for ACF manufacturing, a 3 mm thick Al plate was placed in front of the epoxy specimens. The curing parameters were compared with the ones irradiated without an Al plate. An experiment to bond two PCBs with a pitch of 1000 μm by using a 10:1 mixture of epoxy and conductive particles of 10 μm in diameter was performed by irradiating an EB with an 80 kGy dose under a constant bonding pressure of 196 kPa. DSC measurements were carried out by using a differential scanning calorimeter (TA Instruments, DSC2010) at a heating rate of 10 °C/min from −40 to 150 °C. Approximately 1.6 mg of an EB cured acrylate epoxy specimen was taken and placed in an aluminum DSC pan. The degree of curing was then examined by determining the glass transition temperature of the samples. Raman spectra were recorded using a wide area illumination (WAI) Raman scheme (PhAT System, Kaiser Optical Systems, Ann Arbor, MI, USA). By illuminating a laser (λ_{ex} = 785 nm) into a spot 6 mm in diameter (total illumination area: 28.3 mm^2), the back-scattered light was collected to obtain the Raman spectra of the samples. The sample surface was positioned at the focal point of the WAI scheme and triplicate spectra were collected (laser exposure time 2 min) for each sample [7,8]. For the collection of triplicate spectra, three different spots on the sample surface were randomly illuminated. After averaging of the triplicate spectra, the Raman spectra (spectral range: 1780–600 cm^{-1}) of the samples were linearly corrected at eight different wave numbers (1780, 1680, 1540, 1425, 1165, 975, 715 and 600 cm^{-1}) and then normalized. All the spectral pre-processing including the baseline correction and normalization were performed using MATLAB® version 7.6 (Math Works Inc., MA, USA). The curing behavior of the acrylate epoxy oligomers was analyzed by observing the changes in the intensity of the C=C stretching vibration band of the acrylate monomers at 1635 cm^{-1}. The morphology of the fracture surfaces was observed by using a scanning electron microscope (FE-SEM S-4800 Hitachi Co. Ltd., Japan) at an accelerating voltage of 1 kV. The samples were measured without further treatment.

3. Results and discussion

After EB irradiation, the color of the transparent epoxy samples became brown while the color strength of the glass substrate was proportional to the EB dose. It should be noted that the change of color from transparent to brown after irradiation is a result of the light reflection from the glass substrate, not from the epoxy resin.

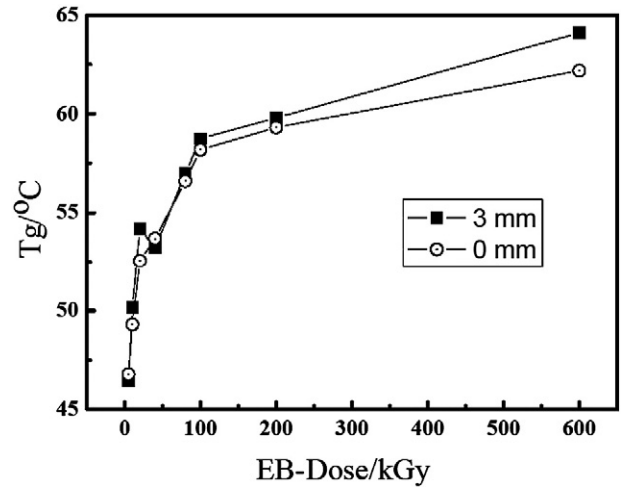

Fig. 2. Plots of glass transition temperatures for E-beam-irradiated acrylate epoxy with and without an Al plate as a function of E-beam dosage. The solid lines are drawn for guide of a reader's eyes.

The influence of the EB irradiation on the curing degree was studied systematically by determining the glass transition temperature (T_g) from DSC measurements, by measuring the gel fraction, and by analyzing the Raman spectra and the SEM images.

DSC measurements offer a simple method of characterizing the curing degree of the EB irradiated acrylate epoxy specimens by determining their glass transition temperatures [9,10]. In Fig. 2, the glass transition temperatures of the EB irradiated acrylate epoxy samples with and without a 3 mm thick Al plate were presented as a function of the EB dosage. The T_g of the samples increased from 46.8 to 62.2 °C for the epoxy composites without an Al plate and from 46.4 to 64.1 °C for their counterparts with a 3 mm thick Al plate with increasing EB dose. The T_g approached 80% of its maximum value at a low EB dose (40 kGy), which is an advantageous property of the EB irradiation for application to ACFs.

Insoluble fractions, such as cross-linked fractions or networked polymers, can be conveniently measured using a gel-fraction determination method [11]. The soluble fractions of the epoxy acrylate oligomers that are not cured during EB irradiation can be dissolved in acetone and removed out of the paper mesh during soaking. Fig. 3 shows the gel fraction values of the EB cured epoxy acrylate oligomers with and without an Al plate according to the dosage of

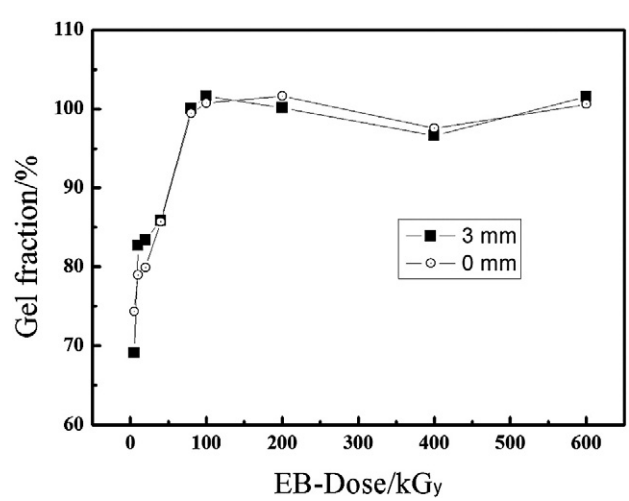

Fig. 3. Plots of a gel fraction for EB irradiated acrylate epoxies with and without an Al plate as a function of EB dosage. The data points are connected by solid lines for guide of a reader's eyes.

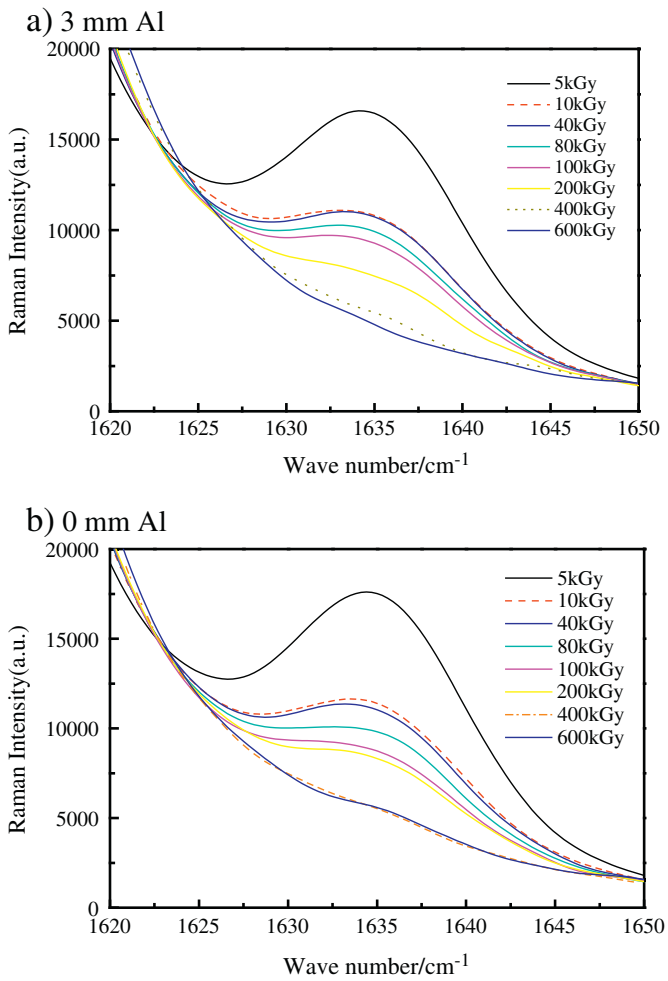

Fig. 4. Plots of Raman spectra for an EB irradiated acrylate epoxy at around 1635 cm^{-1} (a) with and (b) without a 3 mm Al plate as a function of EB dosage.

the EB. The gel fraction values had already reached their maximum at an EB dose of 80 kGy. However, no significant difference was observed between the two samples.

Since the C═C bonds in the functional monomers take part in a cross-linking reaction by radiation polymerization, the absorption band at 810 cm^{-1} related to the C═C twisting vibration of the acrylate groups and the absorption band at 1635 cm^{-1} of the epoxy acrylate oligomers were reduced after UV or EB irradiation [12]. To compensate for the thickness effect of the EB cured acrylate epoxy samples, the C═O stretching band of the acrylate monomers (at around 1730 cm^{-1}) was referred to as an internal standard band. All the spectral intensities of the samples were normalized to the intensity of the sample irradiated with a 5 kGy EB dose. The relative concentration of C═C bonds in EB cured epoxy acrylate oligomers was analyzed by observing the changes in the intensities of the Raman peak at 1635 cm^{-1}. The Raman spectra of the EB irradiated acrylate epoxies covered with and without an Al plate as a function of the EB dosage are shown in Fig. 4. In general, the relative concentrations of the C═C bonds decreased as the dose of the EB irradiation was increased for both cases, while the saturation of curing has occurred at 600 kGy for the epoxy samples with an Al plate and at 400 kGy for the ones without an Al plate. It should be noted that the peaks for the C═C stretching vibration band of the acrylate monomers are superposed on the onset of the next absorption band so that the Raman spectrum of the C═C stretching vibration band appears to change even at the highest EB dose. However, the contribution of the C═C stretching vibration band can be clearly distinguished from that of the neighboring band. This enables it to be analyzed independently.

Scanning electron microscopic images of the fracture surfaces taken from the EB cured acrylate epoxy samples exhibited prominent fracture stripes that illustrate the enhancement of sample toughness [13]. As shown in Fig. 5, the surface morphologies of the samples irradiated with and without an Al plate show no significant difference. This demonstrates that the results from the DSC and the gel fraction measurements are valid.

4. Conclusions

The curing degrees of bisphenol-A type acrylate oligomer irradiated by an electron beam with and without an Al plate (3 mm thick) were investigated by means of DSC, a gel fraction test, Raman spectroscopy, and SEM images. The values of the T_g and the gel fraction of the epoxy resin increased with increasing irradiation dose. In both cases of irradiated with and without an Al plate, the curing degree approached its maximum at a low EB dosage (40 kGy). Furthermore, we bonded two PCB plates using a mixture of an epoxy and conductive particles with an EB dosage of 80 kGy. These results confirm that the investigated epoxy oligomer is suitable for the application to anisotropic conductive films with respect to the reduced bonding process time at an ambient curing temperature.

Acknowledgments

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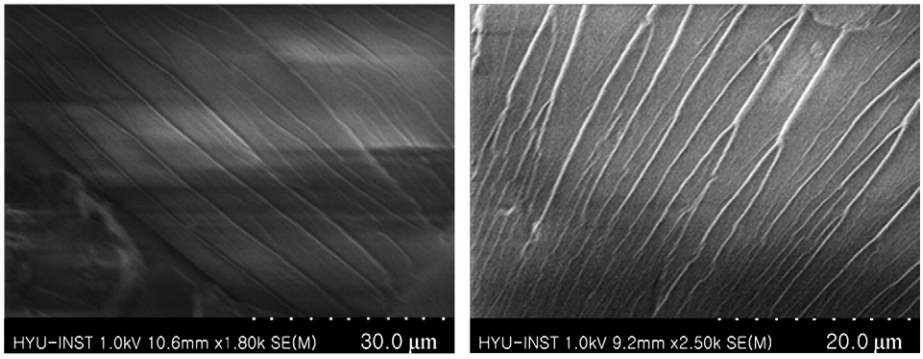

Fig. 5. Representative SEM micrographs of 20 kGy EB irradiated acrylated epoxies (left) with and (right) without placing an Al plate in front of the sample.

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